

Lecture 3:

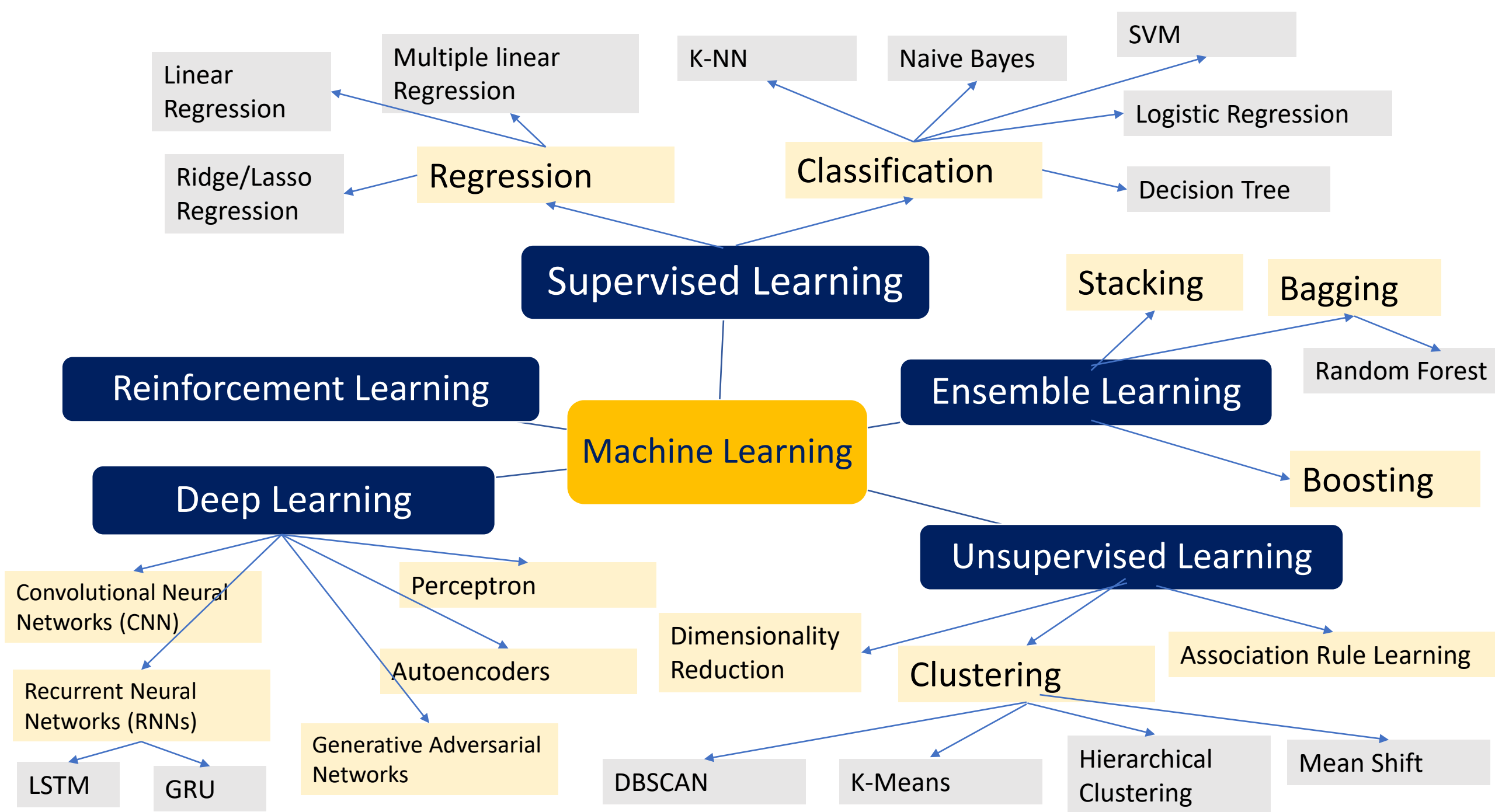
Introduction to Machine Learning and Reinforcement Learning

ML/AI for Engineering Applications
ENV 6932

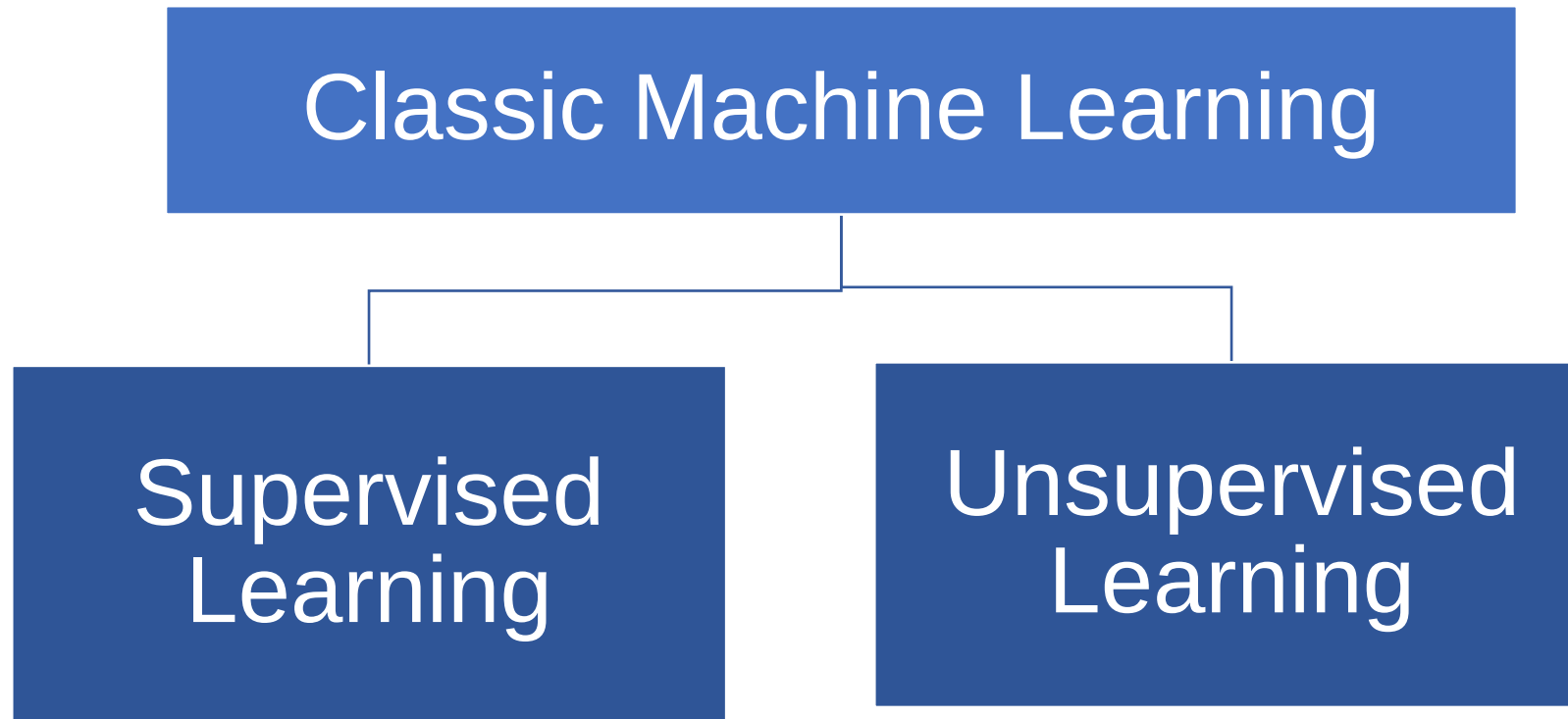
Sara Behdad
Fall 2025

Outline

- Structured vs. Unstructured Data
- Semi-Supervised Learning
- Reinforcement Learning



Two Major Groups of Machine Learning Techniques



Supervised vs. Unsupervised Learning

Supervised Learning	Unsupervised Learning
• Classification: Classifies labeled data • Regression: Predicts trends using previous labeled data • Has more evaluation methods than unsupervised learning • Controlled environment	• Clustering: Finds patterns and groupings from unlabeled data • Has fewer evaluation methods than supervised learning • Less controlled environment

Difference Between Supervised and Unsupervised Learning

- The difference is the **problem statement**.
- **Supervised learning:** we have two types of variables: (1) dependent variable or target variable (y), and (2) independent variables or explanatory variables (x)
- **Unsupervised models:** there is no split between target and independent variables. We aim to group data points by analyzing their similarity.

Structured vs. Unstructured Data

- **Structured Data:** usually quantitative data, organized data, and easily identifiable by ML algorithms. Structured query language (SQL) was developed by IBM (1974) to handle structured data.
- **Tools:** OLAP, MySQL, etc.
- **Use cases:** Customer Relationship Management, Online Booking, Accounting

Pros	Cons
Easily used by ML algorithms	Limited usage (only for intended purpose)
Easily used by business users	Limited storage options
Accessible by more tools	

Structured vs. Unstructured Data

- **Unstructured Data:** usually qualitative data, cannot be processed via conventional data tools, best managed via non-relational (NoSQL) databases and data lakes.
- **Tools:** Hadoop, Azure, Mongo DB.
- **Use cases:** Data Mining, Predictive Data analytics, Chatbots

Pros	Cons
Native format	Requires expertise
Fast accumulation rates	Specialized tools
Data lake storage (pay-as-you-use pricing)	

Key differences between structured and unstructured data

	Structured Data	Unstructured
Sources	GPS sensors, online forms, network logs, web server logs	email messages, word-processing documents, PDFs
Forms	numbers and values	text files, audio, and video files
Models	a predefined data model	stored in its native format
Storage	stored in tabular formats, e.g., excel or SQL databases, less storage space, scalable storage	stored as media files or NoSQL databases, need more space
Uses	ML	natural language processing (NLP) and text mining

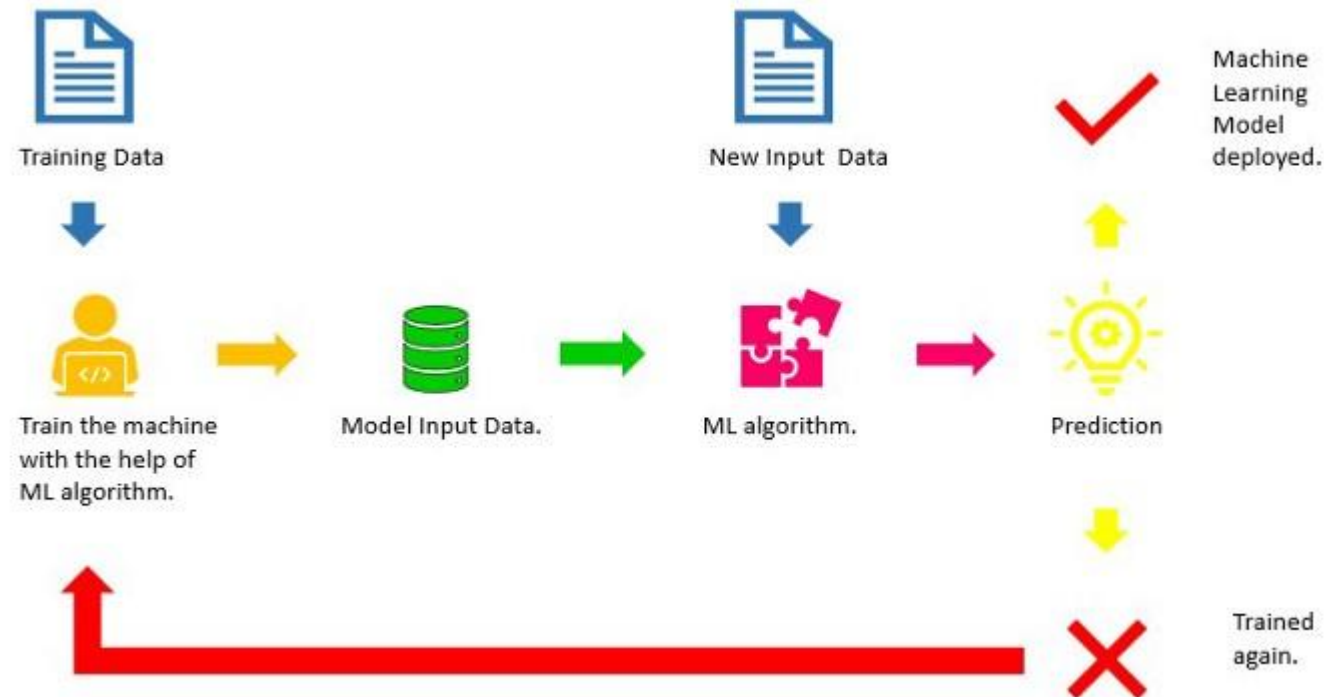
Labeled vs Unlabeled data

- Labeled: data which comes with labels
- Unlabeled: data which does not come with labels

LABELLED DATA		UNLABELED DATA	
			
CAT	ELEPHANT		
			
COW	DOG		

Working of machine learning

- The machine is trained with the help of (1) ML algorithm and (2) training dataset.

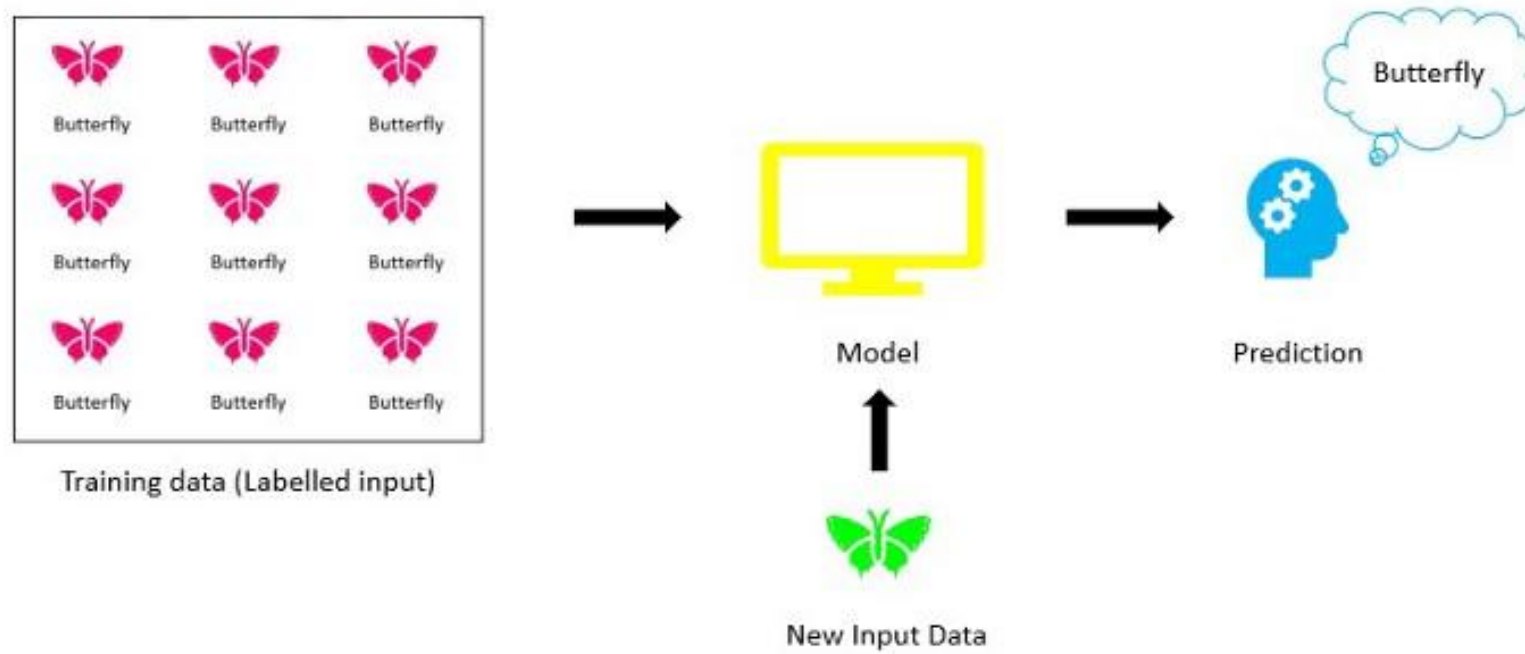


Types of Machine Learning Algorithms

- Supervised Learning
- Unsupervised Learning
- Semi-Supervised Learning
- Reinforcement Learning
- Ensemble learning

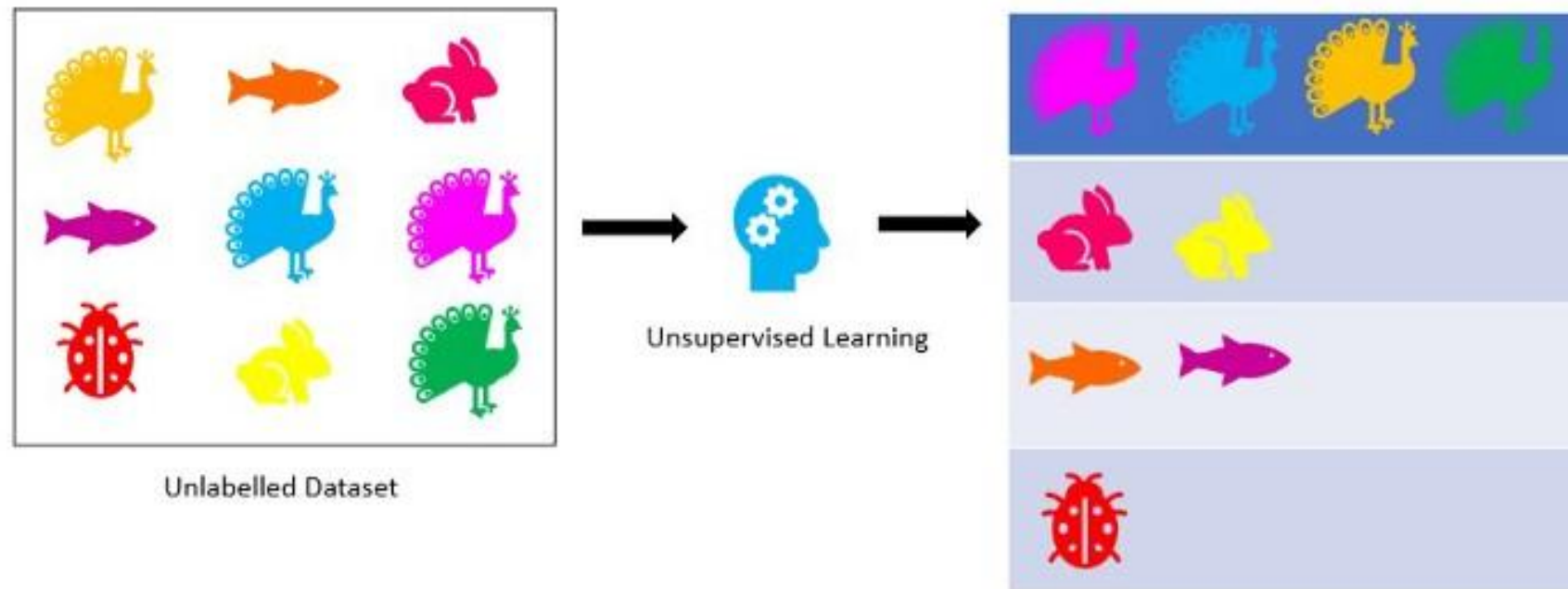
Supervised Learning

- The machine is trained using the labeled data, data already has annotation.



Unsupervised Learning

- The machine is trained using **unlabeled data**.



Semi-Supervised Learning

- Disadvantage of supervised learning: manual labeling of data is a very *costly*, especially with large volumes of data.
- Disadvantage of unsupervised learning: limited applications
- Let's combine them and use semi-supervised learning.
- **How it works?**
 - The machine uses both labeled and unlabeled dataset for training.
 - Similar data is clustered using unsupervised learning,
 - Then, the existing labeled data is used to label the rest of the unlabeled data.

Reinforcement Learning

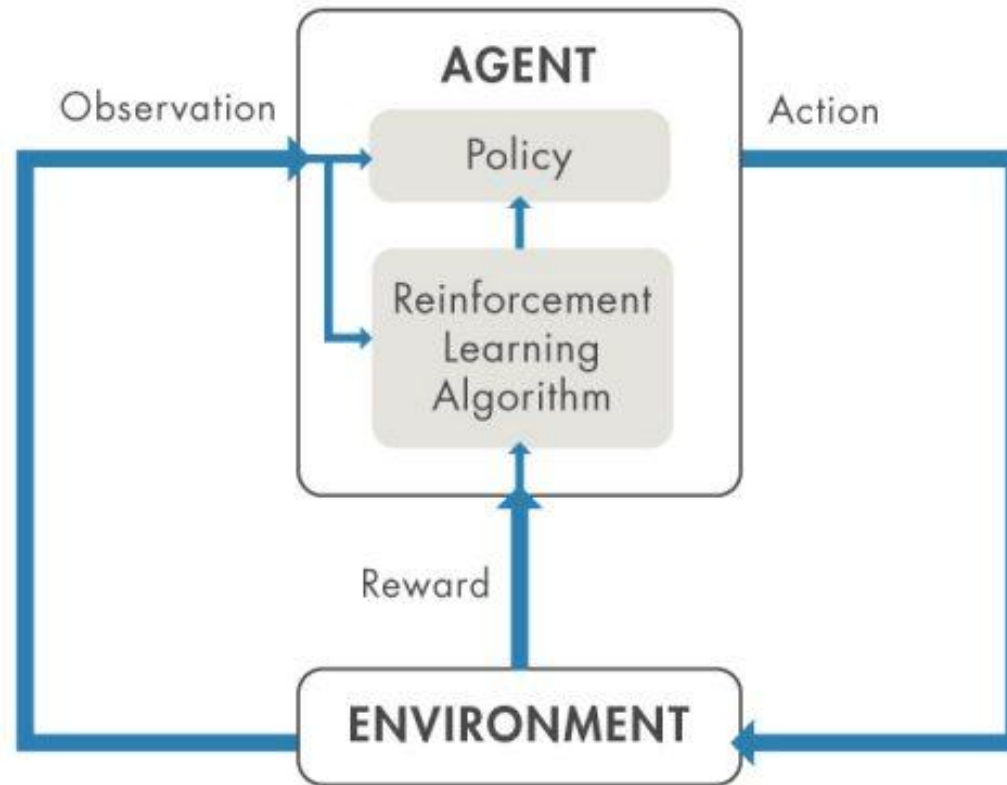
- The machine learns from its experience.
- **Real-life example:** a child in a room who touches a fireplace, gets hurt, and understands that fire is a bad thing when he is too close.
- Main concept: the machine learns from the surrounding by interacting with it.
- **What are common application areas of reinforcement learning?**

Reinforcement Learning

- The machine learns from its experience.
- A real life example: a child in a room who touches a fireplace, gets hurt, and understands that fire is a bad thing when he is too close.
- Main concept: the machine learns from the surrounding by interacting with it.
- **What are common application areas of reinforcement learning?**

Autonomous driving, Robotics, Game Playing

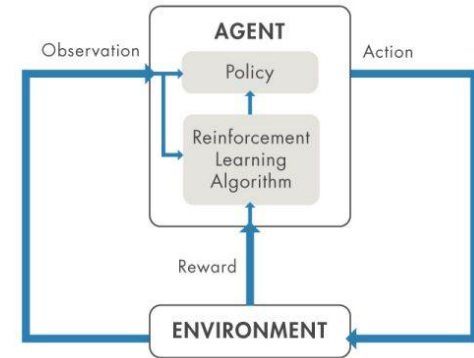
Training an Agent Using RL



Training an agent using RL

It consists of five main steps:

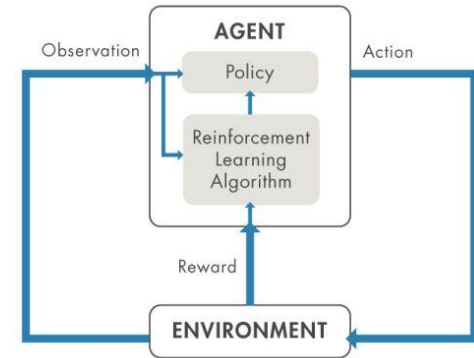
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 - Either a simulation model (usually a good first step), or
 - a real physical system



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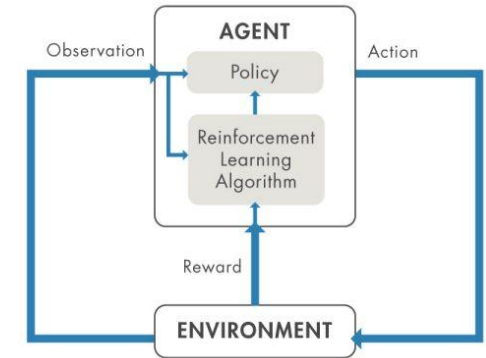
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 - The agent uses reward function to measure its performance.



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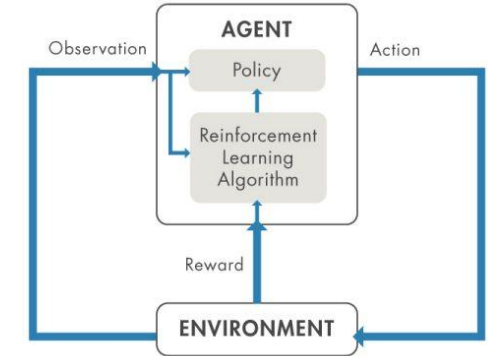
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 - Policy (e.g. represent the policy by using neural networks or lookup tables).
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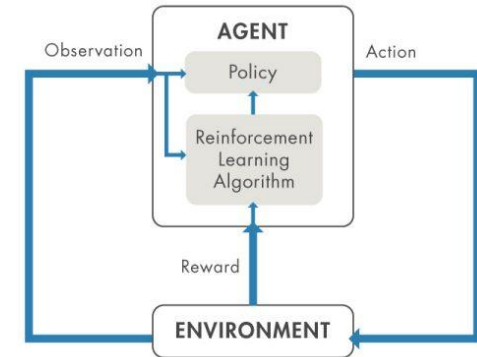
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4. Train and validate the agent
 - Set up stopping criteria, train the agent to tune the policy (e.g. simulation is the easiest way).



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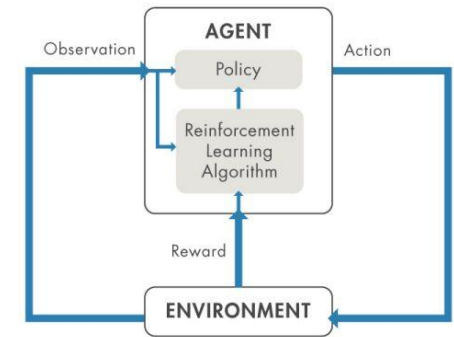
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5. Deploy the policy.



Definition of Policy

- **Concept:** intelligence is an evolving property of the interaction between an agent and its surroundings.
- Analogously, **intelligence is the capacity of an agent to select the appropriate strategy based on its goals.**
- Strategy: a subset of all possible behaviors, “policy”.



Ref: <https://www.mathworks.com/>

Mathematical Definition of Policy

- Let's define a policy π in terms of its Markov Decision Process.
- A Markov Decision Process is a tuple of the form $(\mathcal{S}, \mathcal{A}, P, R)$ structured as follows.
- $\mathcal{S}$ reflects the set of internal states of the agent (e.g., position of a car)
- $\mathcal{A}$: set of the actions of the agent (e.g., possible behaviors that the agent can take in relation to the environment.
- An action can lead to the modification of the state of the agent. This is represented by the matrix P

Mathematical Definition of Policy

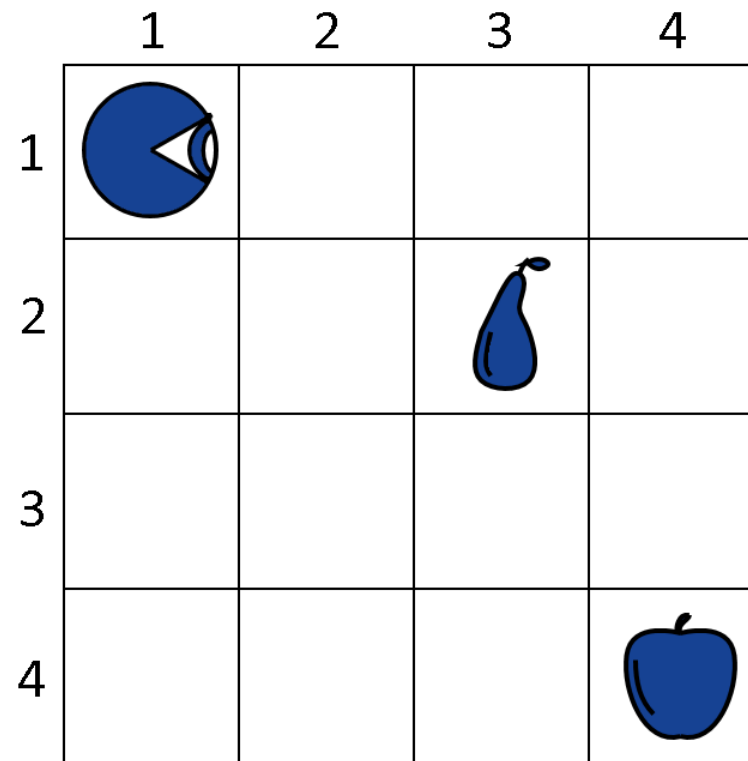
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- Matrix P : the probability of transition from one state to another. Its elements, $P_a(s, \dot{s})$, contain the probabilities $\Pr(s|\dot{s}, a)$ for all possible actions a in $\mathcal{A}$ and pairs of states $(s, \dot{s})$.
- $R(s)$: the reward function for the agent. It takes the state of the agent as input and outputs a real number.

Mathematical Definition of Policy

- Let's formally define the policy, $\pi(s)$.
- $\pi(s)$: The suggested actions that the agent should take for every possible state s in $\mathcal{S}$

Example of a Policy in RL

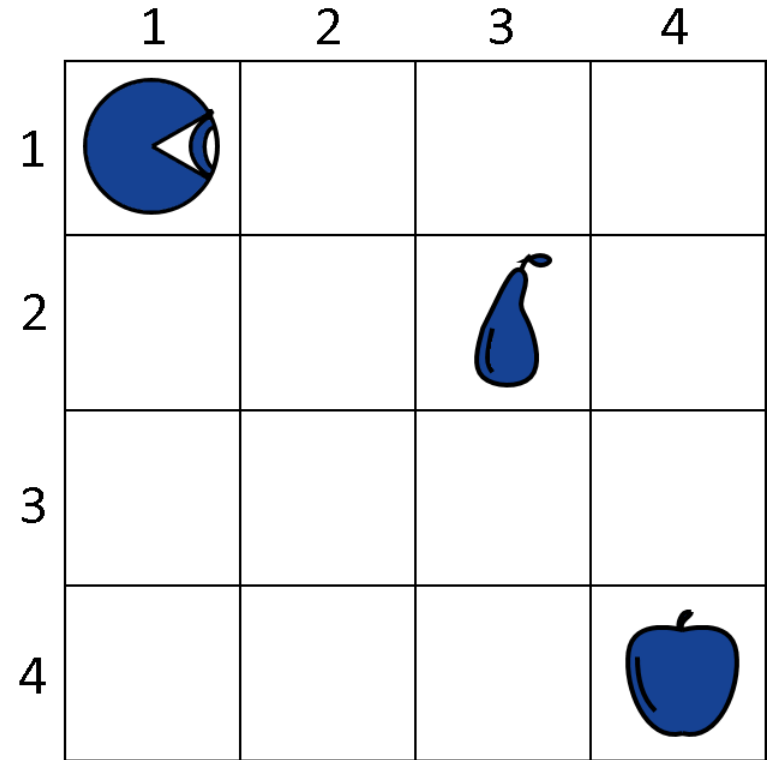
- An agent should search for food from the environment to satisfy its hunger.



Example of a Policy in RL

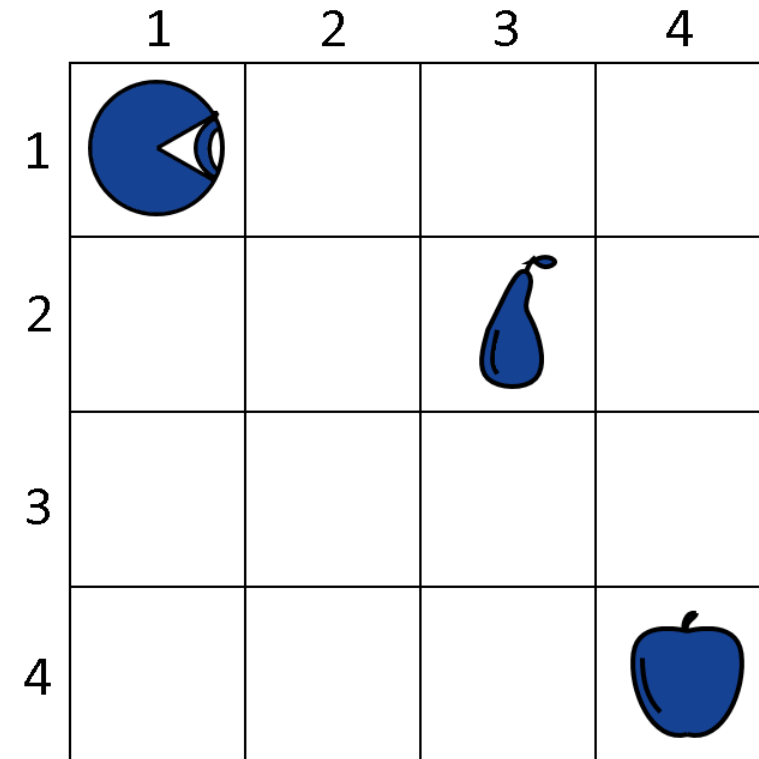
- The agent state: its location on the board, $s_t = (x, y)$ and $s_0 = (1,1)$.
- Action: $A = \{up, down, left, right\}$.
- The probability matrix P contains all pairwise combinations of states (s, s') for all actions in A .

$$P_{\text{down}}((1, 1), (1, 2)) = 1; P_{\text{down}}((1, 1), (1, 3)) = 0; \dots; P_{\text{up}}((4, 4), (4, 3)) = 1$$



Example of a Policy in RL

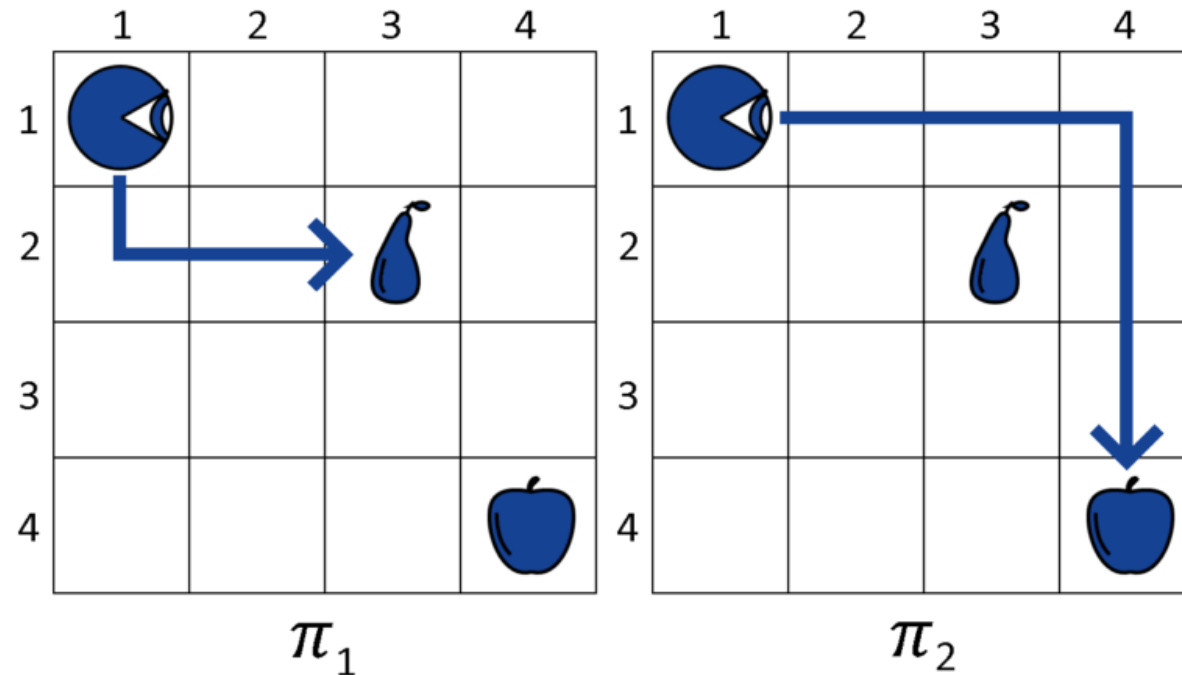
- Reward function:
 - An empty cell, the agent receives a negative reward of -1.
 - If the agent is in a cell with fruit, (3,2) for the pear and (4,4) for the apple, it then receives a reward of +5 and +10, respectively.
 - $R(\text{No fruit}) = -1$
 - $R(\text{Pear}) = +5$
 - $R(\text{Apple}) = +10$



Evaluation of the policies

- Two policies:

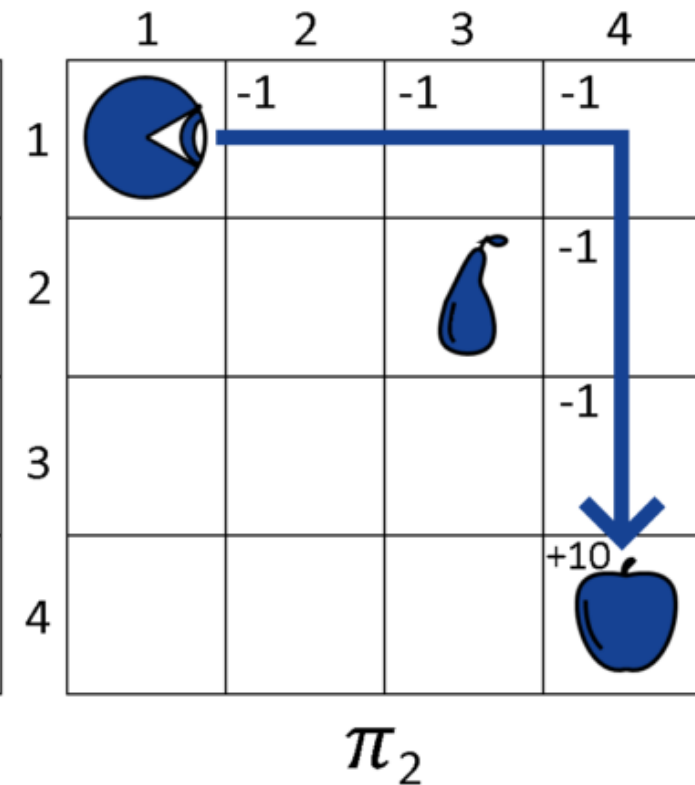
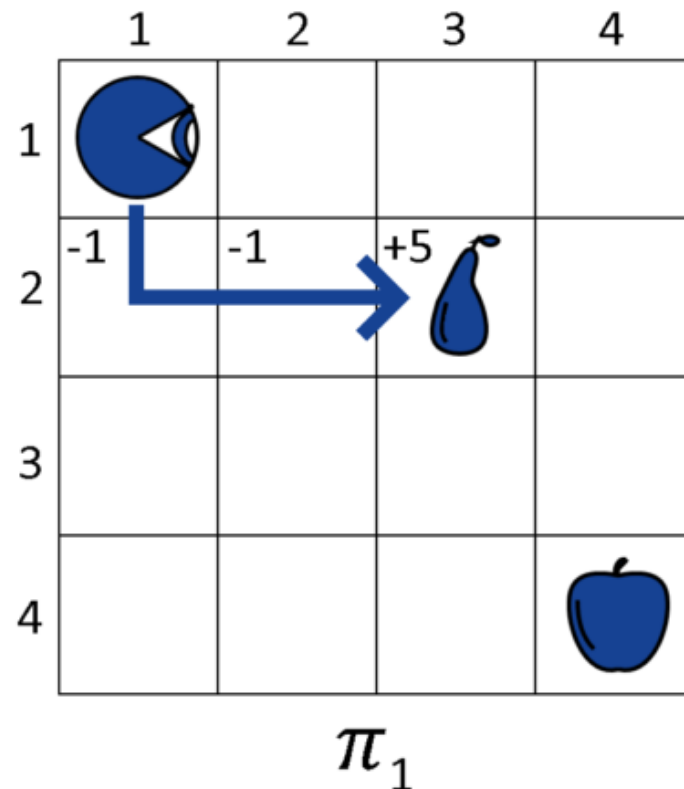
1. π_1 = down, right, right $\rightarrow$ Pear
2. π_2 = right, right, right, down, down, down $\rightarrow$ Apple



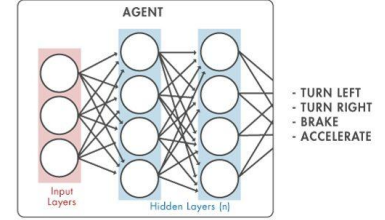
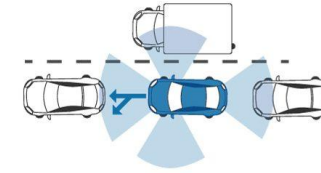
Evaluation of the policies

- The agent needs to select between two policies:

- $U(\pi_1) = -1 - 1 + 5 = +3$
- $U(\pi_2) = -1 - 1 - 1 - 1 - 1 + 10 = +5$

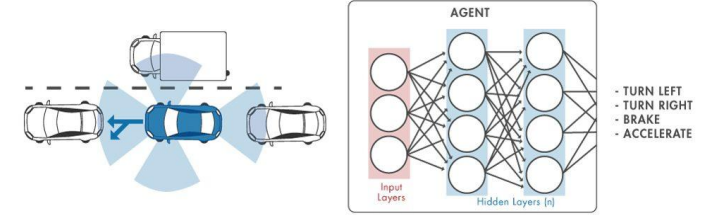


Example: Autonomous parking



- **Goal:** the vehicle computer (*agent*) to position the car in the correct parking spot.
- **Agent:** a neural network that decide how to turn the steering wheel by looking at input from multiple sensors (e.g., camera frames and lidar).
- **Environment:** everything outside the agent (e.g., nearby vehicles, weather conditions, etc.)
- **Actions:** steering, braking, and acceleration commands

Example: Autonomous parking



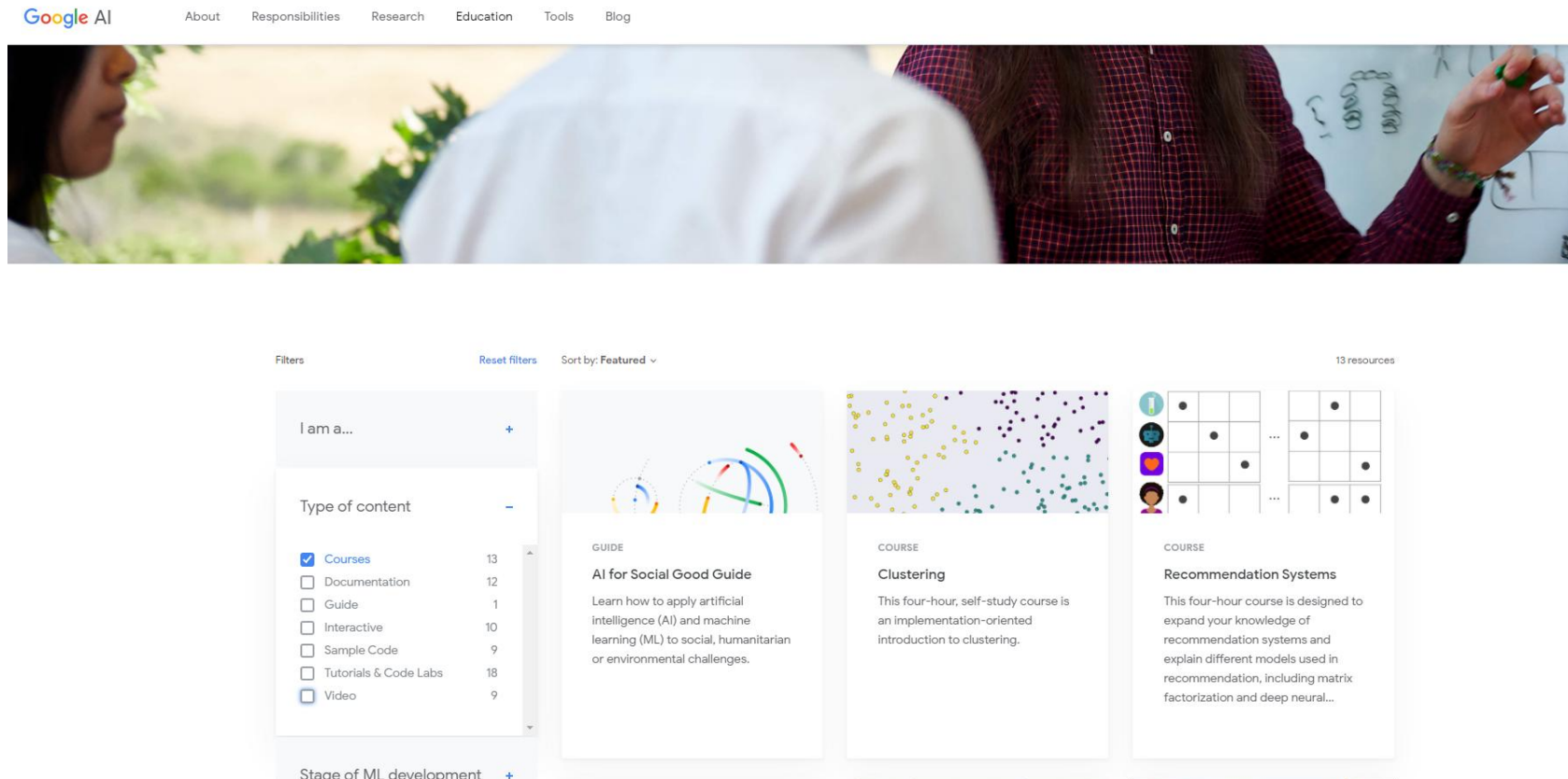
- During training, the agent uses readings from sensors to generate actions. To generate the correct actions (*policy tuning*), the agent repeatedly tries to park the vehicle using trial and error.
- The correct action is rewarded (reinforced) with a numerical signal.

About term paper

- Make yourself familiar with Google AI
- Install Anaconda

About term paper

- Make yourself familiar with Google AI



<https://ai.google/education/>

Data sources for term paper

- Google AI > Using AI for social good > Data Sources
- [Dataset Search](#)
- [Google Earth Engine Catalog](#)
- [Kaggle](#)

The Kaggle logo, featuring the word "kaggle" in a lowercase, blue, sans-serif font.

Kaggle Datasets

Explore, execute, share, and comment on code for any open dataset with the in-browser analytics tool, Kaggle Kernels. You can also download datasets in an easy-to-read format.



Google Earth Engine Catalog

Earth Engine's public data archive includes more than forty years of historical imagery and Earth science datasets, updated and expanded daily.

The Google Dataset Search interface, showing the Google logo, the text "Dataset Search", and a search bar.

Dataset Search

Google's Dataset Search tool allows users to search across thousands of dataset repositories on the Web. Once you've found a relevant dataset, Dataset Search will direct you to the repository ...

Teachable Machine

AboutFAQGet Started

How do I use it?

Class 1

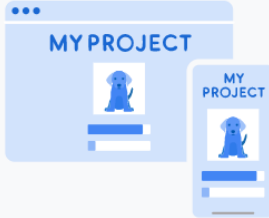


Class 2



TRAIN MODEL





- 1 Gather**

Gather and group your examples into classes, or categories, that you want the computer to learn.

[Video: Gather samples](#) 
- 2 Train**

Train your model, then instantly test it out to see whether it can correctly classify new examples.

[Video: Train your model](#) 
- 3 Export**

Export your model for your projects: sites, apps, and more. You can download your model or host it online for free.

[Video: Export your model](#) 

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